

OCEAN-SHIELD — MARITIME ENVIRONMENTAL ANALYTICS

SIH26143 PROTOTYPE • ANALYST-REVIEW CASE SUMMARY

AUTOMATED SCREENING OUTPUT — NOT A STATUTORY NOTICE OR FINDING OF LIABILITY
Requires calibrated imagery, source provenance, qualified analyst review, and competent-authority action.

Dossier Ref:	ICG-MEP-DOSSIER-20261003-GULF	Date of Issue:	03 Oct 2026, 17:37 UTC
Maritime Sector:	Gujarat Offshore / Northern Arabian Sea	Review status:	SCREENING LEAD — REVIEW REQUIRED
Target Vessel:	MT NEPTUNE GLORY	IMO / MMSI:	9384722 / 636019482
Flag State:	Liberia	Attribution rank:	40.0/100 lead-priority score

1. SATELLITE RADAR REMOTE SENSING (SAR) SCREENING

Sensor Platform:	Sentinel-1A C-Band SAR	Acquisition Timestamp:	2026-09-04 13:10:00 UTC
Swath Mode & Pol:	Interferometric Wide (IW) Swath / VV (Co-polarization optimal for sea surface roughness)	Ground Resolution:	10.0 meters/pixel
Observed Slick Centroid:	22.5845° N, 69.1861° E	Total Spill Surface Area:	0.62 km²
Mass / thickness:	Not estimated from this SAR scene	Geometry screen:	90.0/100 uncalibrated geometry screen
ADIOS Weathering State:	Not assessed: oil profile and mass are required	Evaporative Loss / Mousse:	N/A% Evaporated / N/A% Water Uptake

2. CONDITIONAL HYDRODYNAMIC TRANSPORT SCENARIO

Using an advection-diffusion Lagrangian numerical formulation with the supplied met-ocean field ($\Delta u=0.28$ m/s, $\Delta v=0.18$ m/s) and Ekman surface windage (3.2% Stokes drift factor with 15° Coriolis deflection), the slick plume trajectory was hindcasted backwards in time. The model propagates the supplied **age hypothesis** backward through the validated input field. It does not infer a release time or establish a source location; uncertainty must be quantified against authoritative current, wind, and imagery inputs before operational use:

Conditional backtracked point:	22.38500° N, 69.11500° E	Assumed slick age:	Not supplied
Conditional time offset:	T – 10.5h (not an inferred discharge time)	Total Hydrodynamic Drift:	14.80 km displacement

3. AIS CORRIDOR SCREENING & NAVIGATION CONTEXT

Analysis of **6 vessels** operating in the sector filtered down to **3 corridor candidates**. Vessel **MT NEPTUNE GLORY** is the highest-priority review lead based solely on spatio-temporal proximity. This is not an allegation or responsibility finding:

Evaluation Metric	Observed Telemetry Value	Review Context	Screen Value
Closest Point of Approach (CPA)	7.22 NM (13.37 km)	Modelled proximity to candidate origin	0.0/100
Temporal Coincidence	$\Delta t = 0.00$ hours from origin	Modelled temporal alignment	100.0/100
Speed Over Ground (SOG)	Drop: 14.7 → 5.1 kts	Large speed change (14.7 -> 5.1 kts); requires navigation-context review	70.0/100
Vessel class	Crude / Product Tanker (DWT: 158000)	Descriptive metadata only; excluded from lead score	Not scored
TOPSIS Closeness (C_p)	98.7% (Rank #1)	MCDA 5-criteria Euclidean distance to positive ideal	3 Borda pts
AIS Integrity / Spoofing	Verified Continuous	Gap: 240.0m Accel: 0.14 kts/m	VERIFIED_CONTINUOUS
Bayesian Legal Gate	INSUFFICIENT_EVIDENCE	Shannon Entropy $H_{norm}=0.72$ (Threshold ≤ 0.82) • Prevents false accusation	55.2% Conf

Adversarial Stress Test	FALSIFICATION VULNERABLE (CIRCUMSTANTIAL)	4 physical attacks: ±20% ocean current, ±1% leeway, GPS noise, AIS continuity	25% Robust
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6. COASTAL HAZARD & BEACHING INTERCEPTION ALERT

MODELLED SHORELINE ALERT: Hydrodynamic forecasting estimates possible slick impact along the coastline within 11.2 hours. Interception coordinates: 22.5800° N, 69.4500° E. This scenario output should be reviewed by the responsible incident commander before any response decision.

7. MARPOL 73/78 ANNEX I & SECTION 65B/63 FORENSIC ADMISSIBILITY

MARPOL 73/78 Annex I Parameter	Calculated Telemetry Value	Statutory IMO Threshold (Reg 34/15)	Compliance Verdict
Estimated Slick Oil Volume	739 Liters (~0.7 m³)	Total allowable discharge per voyage: 1/30,000 DWT	Exceeds En-Route Capacity
Instantaneous Rate of Discharge	68.3 Liters / NM	≤ 30 Liters per Nautical Mile (Reg 34.1.c)	CRITICAL VIOLATION
Vessel Operational Speed	5.1 knots	Vessel must be en route (> 0 kts)	En Route Verified
Legal Evidentiary Framework	Sec 65B (IEA 1872) / Sec 63 (BSA 2023)	Digital Evidence Computer System Certificate	Admissible Hash Ledger

8. ANALYST NEXT STEPS & FORENSIC PROTOCOL

- 1. Subpoena certified transponder logbooks and voyage data recorders (VDR) from flag state for MT NEPTUNE GLORY (IMO: 9384722).
- 2. Impound Oil Record Book (Part I - Machinery Space / Part II - Cargo Operations) under MARPOL Annex I Regulation 17/36.
- 3. Match laboratory GC-MS / biomarker finger-printing of physical sea slick samples against vessel bilge / bunker fuel tanks.
- 4. Transmit this dossier and Merkle Root cryptographic digest to the Ministry of Shipping and Indian Coast Guard Maritime Rescue Co-ordination Centre (MRCC).

9. MULTI-ASSET CRYPTOGRAPHIC CHAIN-OF-CUSTODY (MERKLE TREE ROOT)

Forensic Asset Component	Source / System Asset	SHA-256 Cryptographic Hash Digest
SAR Sensor Raster	Sentinel-1 C-SAR	2e83af2e4c51fec670268028de37...a43620ac
AIS Telemetry Broadcast	MMSI: 636019482 (MT NEPTUNE GLORY)	ff05761bf9a61fff1c2e6666c8dc...4d0351ef
Hydrodynamic Met-Ocean Grid	HYCOM NetCDF	7f755a197340b64bc12e04a3bed4...8bf74ab0
PyTorch U-Net Model Weights	models/sar_unet_best.pt	c8105adb50178490c54a3ed818c5...aeb64423
Incident Case Manifest JSON	canonical_manifest.json	1142c78a4b2fb64652bf4a308655...c9b20391

MERKLE ROOT HASH (CHAIN-OF-CUSTODY DIGITAL SEAL): 5eaf8415db169eb0d0d678d19578faaa40d5366828a6ca941fcc20023fe63483 This Merkle Root cryptographically binds the raw SAR imagery, met-ocean current grid, transponder stream, PyTorch U-Net neural weights, and incident manifest into an immutable digital audit trail. Generated in accordance with Section 65B of the Indian Evidence Act, 1872 & Section 63 of the Bharatiya Sakshya Adhiniyam (BSA), 2023.
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